

DL Models for Audio Processing: Speech Recognition

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Audio Main Applications

Last class, we discuss the 3 main applications for audio signals:

- Environmental sounds.
Ex. sound classification, audio tagging, ...
- Music.
Ex. song recognition, musical instrument identification, ...
- Speech.
Ex. voice recognition, speech translation, speaker identification, ...

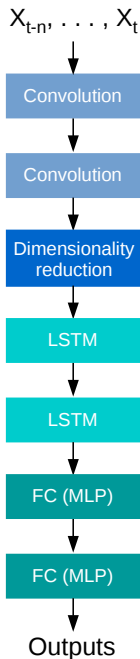
This class focuses on speech applications.

Audio Applications: Speech

- Speech recognition.
- Video captioning.
- Speaker audio separation.
- Language translation.
- ...



- Last class, we reviewed models to analyze environmental audio signals.
- The recommended model is a combination of:
 - CNNs: Learn meaningful local features.
 - RNNs: Learn distance and global temporal features.
 - MLPs: Learn good classifiers.



We can also input raw audio data.
In such a case, we need to include:

- Dilated CNNs
- Aggressive downsampling in the input layer

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Sure you do, but it does not achieve
state-of-the-art performance

Why?

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 - There are almost an infinite number of different sounds.
 - Specific features can be highly relevant to characterize each sound.
 - Learning a suitable set of representative or discriminative features is key.

What About Speech?

What About Speech?

- In the case of speech recognition, feature learning is important but sequence learning is the real key, why?

VOWELS	monophthongs				diphthongs		Phonemic Chart voiced unvoiced	
	i:	ɪ	ʊ	u:	ɪə	eɪ		
	sheep	ship	good	shoot	here	wait		
	e	ə	ɜ:	ɔ:	ʊə	ɔɪ		
CONSONANTS	bed	teacher	bird	door	tourist	boy	show	
	æ	ʌ	ɑ:	ɒ	eə	aɪ	aʊ	
	cat	up	far	on	hair	my	cow	
	p	b	t	d	tʃ	dʒ	k	g
	pea	boat	tea	dog	cheese	June	car	go
	f	v	θ	ð	s	z	ʃ	ʒ
	fly	video	think	this	see	zoo	shall	television
	m	n	ŋ	h	l	r	w	j
	man	now	sing	hat	love	red	wet	yes

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 - For a given language the space of phonemes is not so large.

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	e bed	ə teacher	ɜ: bird	ɔ: door	ʊə tourist	ɔɪ boy			əʊ show
	æ cat	ʌ up	ɑ: far	ɒ on	eə hair	aɪ my			aʊ cow
CONSONANTS	p pea	b boat	t tea	d dog	tʃ cheese	dʒ June	k car	g go	
	f fly	v video	θ think	ð this	s see	z zoo	ʃ shall	ʒ television	
	m man	n now	ŋ sing	h hat	l love	r red	w wet	j yes	

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Decoding speech from an audio signal is not an easy problem



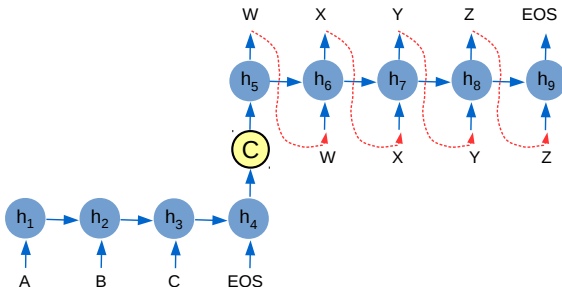
What Type of Model?

- MLPs (Feedforward fully connected NNs)
- CNNs
- RNNs
- GANs
- Seq2Seq (Encoder/Decoder models)
- Seq2Seq with attention
- Transformer and relational models
- Reinforcement learning
- Imitation learning
- Reasoning: Neuro-Symbolic models

Let's Recall Seq2Seq Models

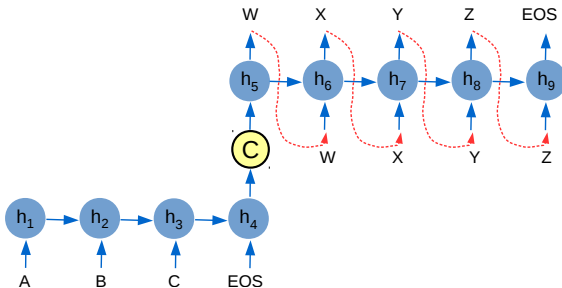
Seq2Seq Model for Language Translation (Sutskever et al., 2014)

- Ex. Textual language translation.
 - Encoder** reads an input sentence (ex. "ABC")
 - Decoder** produces an output sentence (ex. "WXYZ").



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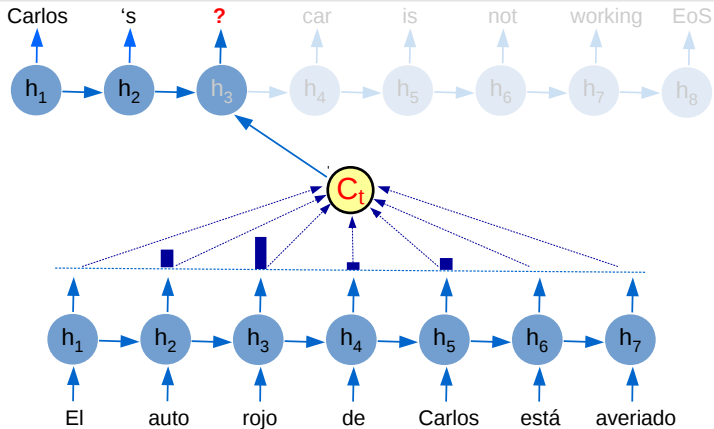
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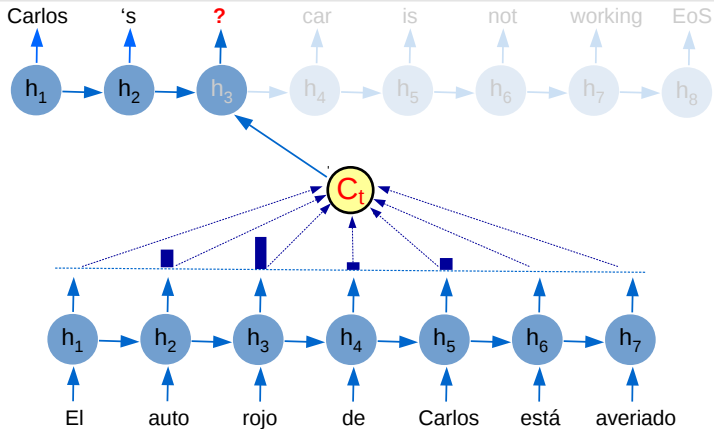
Middle contextual embedding C is **fixed**!

Let's recall Seq2Seq models with attention

Seq2Seq Model with Attention (Bahdanau et al., 2015)



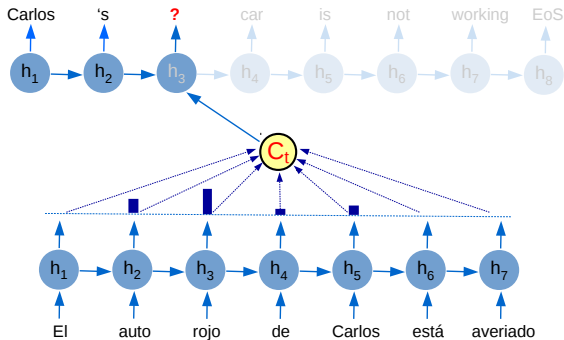
Seq2Seq Model with Attention (Bahdanau et al., 2015)



Middle contextual embedding C_t is **adaptive**!

Can We Apply a Seq2Seq+Att Model to Speech Recognition?

What are the inputs and outputs ?



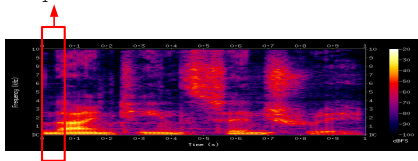
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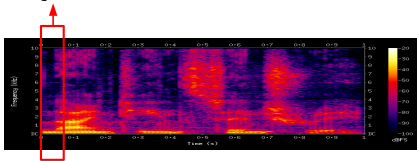
$x_1=40D$



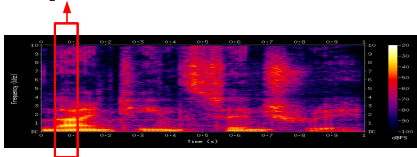
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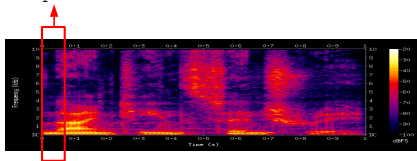
$x_2=40D$



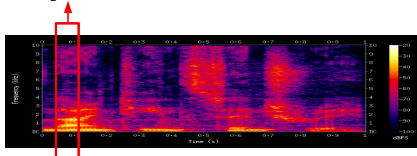
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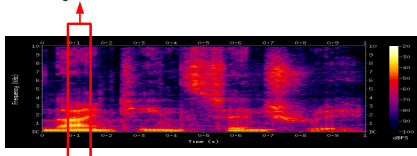
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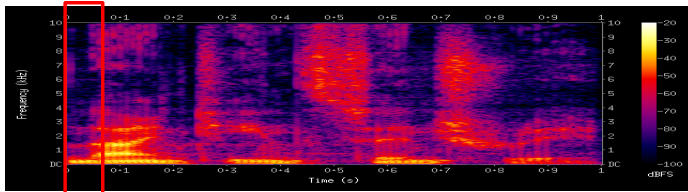
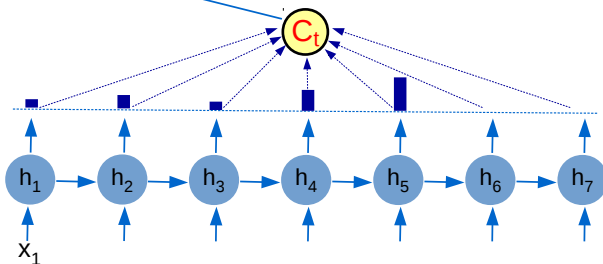
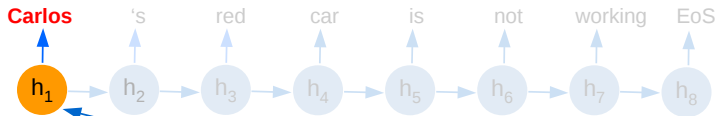


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$x_3=40D$



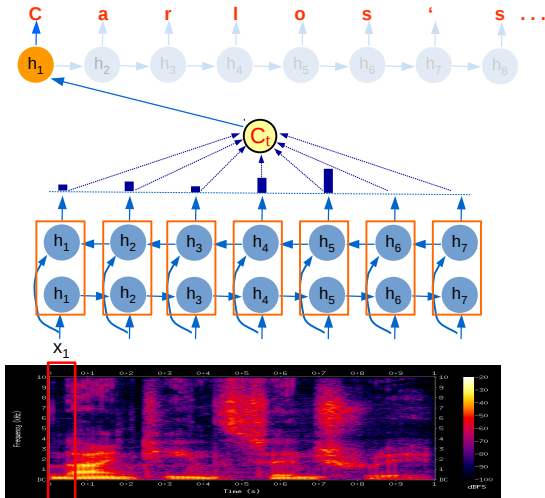


Output Representation

The usual outputs are chars

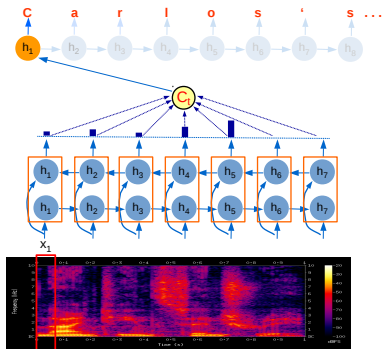
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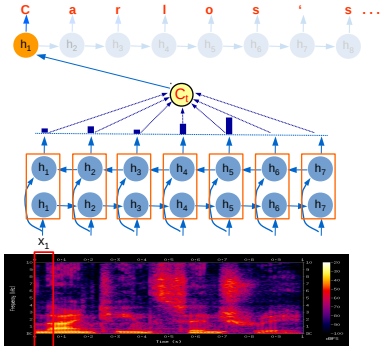
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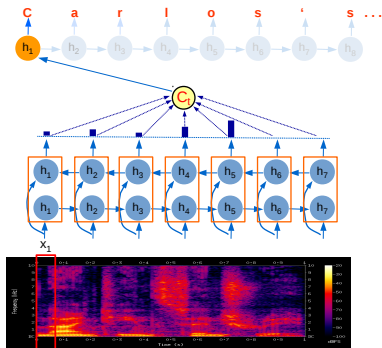
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• "a ... z"; ". "; ", "; " "; "EOS"; "0-9".

Output Representation

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- "a ... z"; "."; ","; " "; "EOS"; "0-9".
- Softmax classification.

Example 1: DL Architecture for Speech Recognition

Input:

- 40D log-mel feats for overlapped segments of 10-20ms. 5-10ms overlap.

Encoder :

- 4 stacked bidirectional-LSTM layers.
- Hidden state 256D.

Decoder :

- 4 stacked bidirectional-LSTM layers.
- Hidden state 256D.

Head

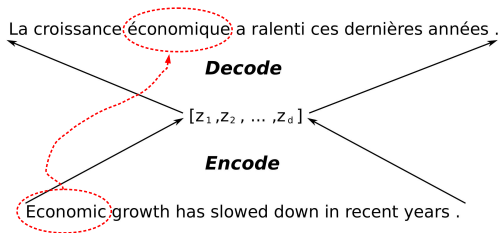
- Character based output. "a ... z"; ". "; ", "; " "; "EOS"; "0-9".
- Activation: soft-max.

Input-Output Segmentation

- One problem with RNNs is that, at each prediction time, they are trained to perform discrete label classifications.

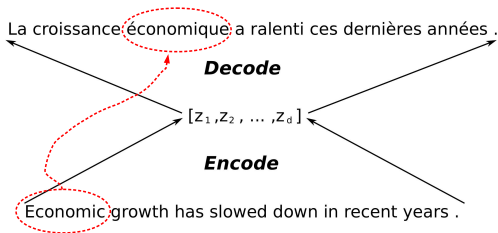
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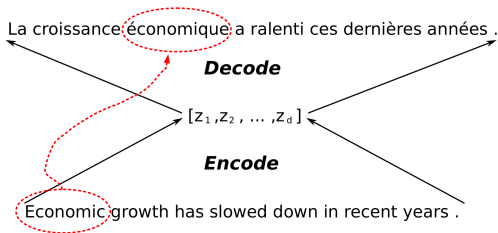
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- In the case of speech recognition signal segmentation is an issue. As an example, in the previous model, outputs correspond to segmented "chars".
- However, inputs producing each output are not pre-segmented. Relevant information is hidden in the Mel features.

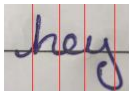
IDEA: Add to the output an **blank** token.

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The use of an **blank** output label facilitates the task of sequence alignment between the input and output, it can be considered as a waiting or pause output symbol.

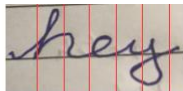
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t0 t1 t2 t3 t4
Image for annotation

-hey-

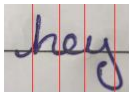


t0 t1 t2 t3 t4 t5 t6
Image for annotation

-hheyy-

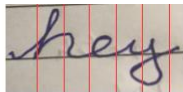
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- Methods such as Connectionist Temporal Classification or CTC (Graves et al., 2006) or Pooling Over Time (Bahdanau et al., 2016) help to deal with the misalignment issue.

CTC, Graves et al., 2006

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CTC network has a softmax output layer with an extra label corresponding to observing an blank output.



We start with an input sequence, like a spectrogram of audio.



The input is fed into an RNN, for example.

h	h	h	h	h	h	h	h	h	h
e	e	e	e	e	e	e	e	e	e
l	l	l	l	l	l	l	l	l	l
o	o	o	o	o	o	o	o	o	o
ε	ε	ε	ε	ε	ε	ε	ε	ε	ε

The network gives $p_t(a | X_t)$, a distribution over the outputs $\{h, e, l, o, \epsilon\}$ for each input step.

h	e	ε	l	l	ε	l	l	o	o
h	h	e	l	l	ε	ε	l	ε	o
ε	e	ε	l	l	ε	ε	l	o	o

With the per time-step output distribution, we compute the probability of different sequences

h	e	l	l	o
e	l	l	o	
h	e	l	o	

By marginalizing over alignments, we get a distribution over outputs.

Example 2: DL Architecture for Speech Recognition

Input:

- 40D log-mel feats for overlapped segments of 10-20ms. 5-10ms overlap.

Encoder :

- 4 stacked bidirectional layers.
- Hidden state 256D.

Decoder :

- 4 stacked bidirectional layers. Pooling in time window size=2.
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Head

- Representation: character based output. "a ... z"; ". "; ", "; " "; "EOS"; "0-9".
- Activation: soft-max, using CTC.

Performance Metric

Word error rate (WER) metric:

$$\text{WER} = \frac{\# \text{Insertion} + \# \text{Deletions} + \# \text{Substitutions}}{\# \text{tokens}}$$

